

Deepak Mallampati

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Kent, OH

PROFESSIONAL SUMMARY

0 → 1 builder shipping production LLM systems end-to-end: Manga Lens (Chrome Web Store, 4 AI vision providers, 29 sites, per-provider failure isolation), Claims Processing (5-stage multi-agent system with schema-validated JSON contracts), Dream Decoder (multimodal text → image). Python + TypeScript full-stack: FastAPI, React/Next.js, Postgres + pgvector, Docker. Production RAG with ~35% retrieval precision at Progress Solutions. Master's in Computer Science, Kent State.

CORE COMPETENCIES

Applied AI / LLMs • RAG + Grounding • Agentic Workflows • Schema Contracts • Eval Harnesses • Python / FastAPI • TypeScript / React • Docker / CI-CD • Multimodal Vision • Predictive ML

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Agent Engineering, Production LLM

USA - Healthcare Technology, AI Consulting

- Built **Retrieval-Augmented Generation (RAG)** for clinical knowledge retrieval; recursive semantic chunking and transformer embeddings improved retrieval precision ~35%, reduced irrelevant retrieval >30%, response alignment >90%.
- Developed **agentic LLM workflows** for multi-step healthcare queries with tool discipline, structured reasoning, and grounding rules; response stability ~25%, hallucinations cut >30%.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show and care engagement scoring; recall +15-20% on high-risk categories, precision held >90% via class weighting, stratified sampling, threshold calibration.
- Packaged ML/LLM inference as **FastAPI / Flask** services on **Docker** with structured logging and load simulation; post-deploy defects ~30%.
- EHR preprocessing (Pandas, NumPy); dataset reliability >98%, downstream instability -40%.
- HIPAA-conscious governance: de-identification, data lineage, evaluation audit trails, stakeholder-facing system-limitation docs.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

Hyderabad, India - Oil & Gas ERP

- Optimized **T-SQL stored procedures** on a compliance-sensitive ERP; query time -20%, retrieval latency -25%.
- Built **Jenkins CI/CD pipelines** for schema updates and stored-procedure deployments; deploy errors -30%, release cycle -35 to -40%.
- Designed performance dashboards with **SQL Server DMVs + Grafana**; incident recurrence -25%, root-cause resolution -30%.
- Tuned index maintenance and partition-aware queries; reconciliation runtime -15%, deadlocks -15%.
- Supported RBAC and audit logging for financial modules in a regulated environment.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer

Hyderabad, India - Nonprofit, Video Analytics, Applied NLP

- Built **transformer-based video summarization** over 5,000+ recorded sessions; hierarchical summarization and timestamp-aligned clip extraction cut manual review 60-70%.
- AI-selected highlights aligned with human-curated moments ~85%.
- Implemented **document Q&A with RAG**; hallucinations ~30%, contextual accuracy >85%.
- Deployed as **Flask** API with a lightweight web interface for non-technical staff.

PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

5-stage multi-agent pipeline (Intake, Validation, Consistency, Duplicate, Risk Scoring) with schema-validated JSON contracts between agents to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policies. Audit-ready explainable risk scoring.
Python, LangChain, FastAPI, pgvector, GPT-4, schema contracts

Manga Lens - Multi-Provider AI Vision Chrome Extension

Real-time manga translation extension shipped to the Chrome Web Store. Multi-provider abstraction (4 AI vision providers) with per-provider payload handling for worst-case failure isolation. 29 sites supported across manifest V3 service workers.
TypeScript, Manifest V3, Service Workers, OpenAI / Anthropic / Google / OpenRouter

Dream Decoder - Multimodal Creative Platform

End-to-end multimodal pipeline: text description, intermediate prompt transformation, grounded image generation. ~30% contextual alignment gain, ~25-30% first-pass image success.
FastAPI, React + Vite, Tailwind, Perplexity Sonar, Replicate

YOLOv8 Driver Drowsiness Detection

Real-time computer vision for driver drowsiness; sub-50ms inference on edge hardware. Walk-forward validation and class-weighted training for imbalanced safety data.
PyTorch, YOLOv8, OpenCV, ONNX

EDUCATION

Master of Science, Computer Science - Kent State University *Aug 2023 - May 2025*
Focus: Applied Machine Learning, NLP, Database Systems.

Bachelor of Technology, Computer Science - KL University *Jun 2019 - May 2023*
Founder / Lead of E-Farming platform (university project, pilot with 3 cooperatives).

CERTIFICATIONS

Deep Learning Specialization - *DeepLearning.AI (Coursera)* *2024*

SKILLS

Languages: Python, TypeScript, JavaScript, SQL (T-SQL, PostgreSQL, SQLite), Java, C++
AI / ML: PyTorch, scikit-learn, XGBoost, LangChain, LangGraph, LlamaIndex, OpenAI, Anthropic, Hugging Face Transformers, Diffusers, YOLOv8
LLM / GenAI: RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, embeddings, vector search
Backend: FastAPI, Flask, REST, WebSockets, Docker, Celery, Playwright
Frontend: React, Next.js, Vite, Tailwind, Chrome Extensions (Manifest V3)
Data: PostgreSQL + pgvector, MongoDB, Redis, Pandas, NumPy, Snowflake
DevOps: Docker, Jenkins, GitHub Actions, Grafana, Prometheus, AWS (EC2, S3, Lambda)
Domains: Healthcare AI (HIPAA-conscious), Applied AI, Enterprise AI, Workflow Automation, Computer Vision, Multimodal